Lesson 2.1: OpenAI API and Anthropic SDK integration in Node.js and Python

What you learn

- **The call.** Send messages (a system message and the user's message) to the model API and read the reply. Both providers have official SDKs for Node.js and Python.
- **API keys.** Keep keys in environment variables on the server, never in front-end code or the repository.
- **Handling errors.** Expect rate limits and timeouts; retry with a short wait and give the user a clear message.
- **Costs.** You pay for input and output tokens; set limits and log usage.
- **Model names.** Names change; read the provider's current model list rather than copying an old name.

Tools and resources (official sites, not checked live)

- Anthropic documentation
- OpenAI documentation

ASSIGNMENT

Practice for lesson 2.1

Business problem: Build a one-route summariser.

Steps

1. Create a server route that accepts text.
2. Call the model with a system instruction.
3. Return the summary and token usage.
4. Add a length limit.

Expected output: A route that returns a short summary and refuses huge inputs.

SOLUTION (ANSWER KEY)

Reject input above your limit before calling the model to control cost.

Check yourself

1. Where does the API key live?
   Answer: In a server environment variable.
2. Why a length limit?
   Answer: To control cost and avoid failures.

Lesson 2.1 of the **AI and Automation** lessons. See this lesson and its official links on skillnavora.in · All chapters and lessons · Open the full lesson guide. Tools, prices and plans change; confirm on each official site.